NATIVE SCALAR / POLICY FOLLOW-UP · FROZEN HISTORICAL APPENDIX

Faster rho batching, with the exact scalar candidate retained

Fresh matched follow-up against the tuned degree-83 control · run-20261012T1543Z

Control: Fold field · reference u64 scalar · 64 lockstep lanes · DP6 · escape off.

Frozen selected policy: reference scalar · Fold field · 256 lockstep lanes · DP6 · escape off.

Five fresh held-out target blocks, three seeds per arm; no historical timing denominator.

139 completed rho processes independently replayed; 0 invalid answers or failed/timed-out solver processes.

The selected 256-lane policy lowers process cost against the tuned 64-lane control on this fresh five-target panel: native current/selected ratio 1.069651 [1.043178, 1.089238]. Mean process cost is 2.987008 s versus 2.780545 s. Exact Barrett arithmetic is verified but unselected. The lower interval edge overlaps the observed A/A ratio range, so the finding remains conditional on this host and panel. The combined edition preserves all prior 60 pages.

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SECTION 01

What changed, and what the new comparison measures

> Native current-tuned / selected whole-process geometric ratio 1.069651 [1.043178, 1.089238]; mean actual process costs 2.987008 s and 2.780545 s. All five target means are lower under the selected reference-scalar Fold256/DP6/off policy.

The optimized control is the prior selected Fold field, 64 lockstep lanes, DP6, escape disabled and reference u64 scalar arithmetic. The scalar candidates instantiate distinct generic implementations in one frozen executable. After training retains reference arithmetic, both held-out arms share that reference-scalar monomorphization and differ only in lane count.

Training freezes one scalar kernel and lane/DP configuration before five fresh held-out points, with three paired seeds each. Primary whole-process wall includes setup through producer replay. All 139 processes independently verify.

Recorded native estimand	Current / selected ratio	95% target-block interval
Primary whole process	1.069651	[1.043178, 1.089238]
Secondary online	1.075203	[1.050382, 1.094289]
Secondary arithmetic mean process	1.074252	[1.048202, 1.090360]

The selected change is a larger lockstep batch: 256 lanes instead of 64, retaining reference scalar arithmetic, Fold field and DP6. Exact Barrett is slightly slower in training and remains unselected. The observed runtime improvement is within the declared grid and host; it does not establish a global optimum or change the generic counting scale. This follow-up runs no new IC campaign and constructs no new IC/rho ratio. Historical matched studies remain intact.

Native descriptive scope	n	Median	Minimum
Current pooled process (s)	15	2.833963	1.152991
Selected pooled process (s)	15	2.850072	0.963513
Paired seed current/selected ratio	15	1.075774	0.994348

The selected pooled median is slightly higher than current, although all five target means improve. Pooled medians and 15 seed-pair ratios are descriptive estimands; they do not replace the primary five-target estimator. Every seed is retained.

The primary point estimate 1.069651 exceeds the maximum A/A ratio 1.046541, while its lower interval edge 1.043178 lies inside that observed noise range. A/A is descriptive, not a population confidence bound; independent-host confirmation remains open.

Sources: N01, N02, N05, N11. Native descriptive supplement.

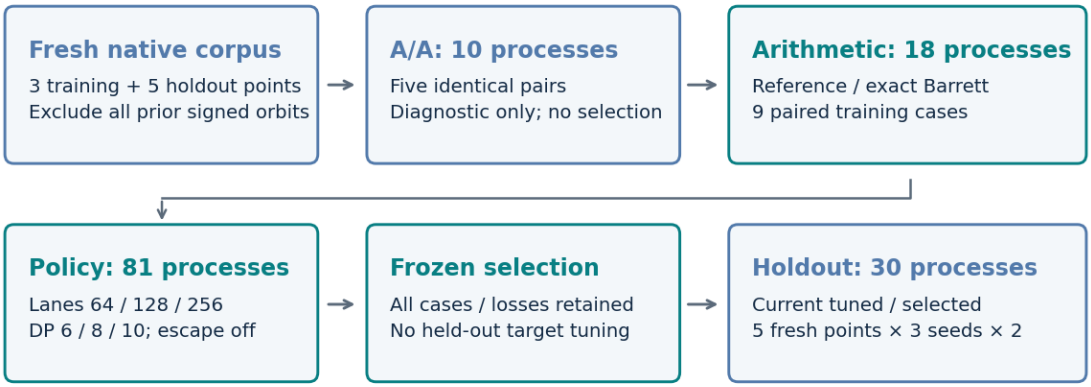
SECTION 02

Exact retained curve and fresh disjoint targets

The model is `icv1-f2m83-t6151469093347-9eba6cd5`: $y^2+xy=x^3+x^2+1$ over GF(2)[z]/(z^83+z^7+z^4+z²+1), subgroup r=8569786107849059, cofactor 1128547018, G=(661424465139818486029201 5,886826041002924939153952 9). Degree 83 describes the field, while the subgroup has about 52.928 bits. The signed Frobenius quotient has 166 phases, with λ=2998004823950457.

Native C++ generation uses seed 202610126201 and conditionally uniform nonzero scalar proposals. It excludes the signed Frobenius orbits of the original exposed target, all K600 points, all eight prior optimization targets, and other accepted new points. These are sampled without replacement under the declared exclusions. A separate validation sidecar is checker-only; each solver receives a point-only file.

Frozen 139-process follow-up: exact arithmetic, policy and fresh holdout



Whole-process comparison against the previously tuned control. No new IC run or IC/rho ratio.
Source: native PROTOCOL.md / stage manifests · run-20261012T1543Z · recorded values only

Frozen stages: A/A10 + arithmetic18 + policy81 + holdout30 = 139 fresh processes. The two selections precede holdout; all losses and failed attempts are retained. [N01, N03–N09]

The fixed native draw stream accepts eight scalar proposals and rejects none. The 609 prior signed orbits cover the original exposed target, K600's 600 representatives and the eight old optimization targets. All eight new points and their orbit exclusions pass independent replay; validation scalars are confined to the checker sidecar. Full points and hashes are in the generator receipt.

Sources: N01, N08, N07, N09.

SECTION 03

Exact Barrett reduction and conditional addition

The candidate replaces the reference hot scalar multiplication's 128-bit remainder with an exact precomputed reciprocal. Its domain is $2 \le m \le 2^{64}-1$ and canonical $a,b < m$. Let $B=2^{128}$, $x=ab$, $\mu=\lfloor B/m \rfloor$ and $q=\lfloor x\mu/B \rfloor$, computed by the exact high 128 bits of a 128×128-bit product. The implementation obtains μ without representing B: $\mu=(B-1) \operatorname{div} m$ plus 1 exactly when $(B-1) \bmod m=m-1$.

Written proof: $B/m-1 < \mu \le B/m$ and $x \ge 0$, so $x/m-x/B \le x\mu/B \le x/m$. Since $x/B<1$, $x/m-1 < x\mu/B$, hence $\lfloor x/m \rfloor-1 \le q \le \lfloor x/m \rfloor$. Thus $0 \le x-qm < 2m$ and a single conditional subtraction returns $x \bmod m$, including $x=0$. Every product, carry and subtraction uses integer arithmetic; no floating-point reciprocal is introduced.

The high product splits each operand into two 64-bit limbs. For $p00=a0b0$, $p01=a0b1$, $p10=a1b0$ and $p11=a1b1$, the middle carry is $(p00>>64)+low64(p01)+low64(p10)$. The upper half is $p11+(p01>>64)+(p10>>64)+(carry>>64)$. The exact limb expansion and finite native oracle checks have separate receipts; neither finite tests nor a polynomial certificate alone proves all machine-word conditions.

For modular addition, add two canonical u64 values with a carry flag. If carry is set or the wrapped sum is at least m, subtract m with wrapping; otherwise retain the sum. Since $a+b < 2m$, this yields the exact canonical residue even when the machine addition carries. The candidate delegates its uniform sampler to the same UniformInt<u64>, preserving the reference random stream.

Verification layer	Actual scope / status
New worker exactness tests	7 release tests pass, including high-product boundaries, arbitrary-width modular arithmetic, strict configs, identical sampler streams and exact fixture solve.
Independent GMP / integer oracle	100,049 high-product checks; 1,048,575 multiply and 1,048,575 add checks over 1,023 moduli; 41,936 products take the correction branch.
Identity certificate	IDC1hc23ae1d80fc5fefa: limb polynomial over $F_{(2^{61}-1)}$, 64 statement-bound + 64 additional points. Finite-field expansion only; proof field null.
Lean mathematical declarations	4 checked Init-only Lean 4.34.1 declarations: limb expansion, actual floor-reciprocal residual bound, one correction, and canonical add. Axiom and fresh-kernel checks pass; Rust carry/casts, MAX reciprocal and wrapping word bridges remain manual/native.

Sources: N02, N10. Written proof is distinct from finite test evidence and formalization status. Lean proof map exposes the remaining machine bridges.

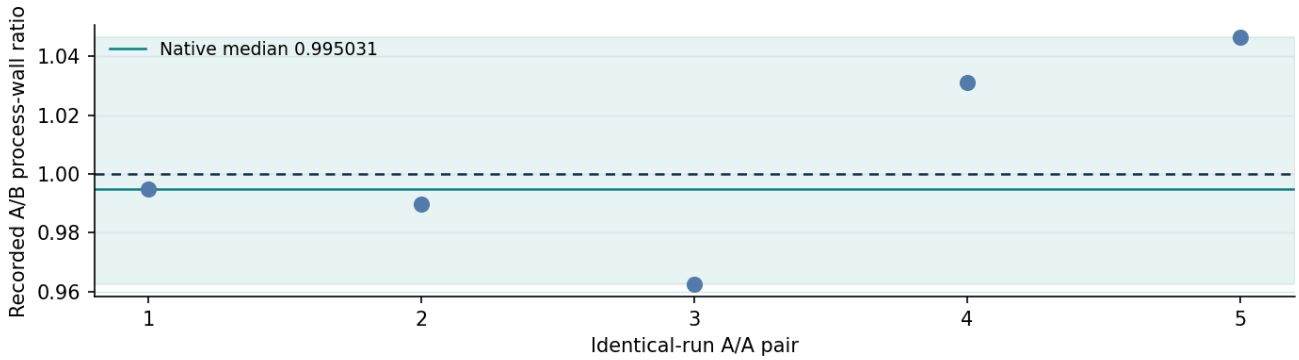
SECTION 04

A/A noise and scalar-kernel selection

Five identical fresh-process reference pairs on training_1 and seed 202610126211 quantify runtime spread. Their answers and exported counters must match. These ten processes do not select a kernel or policy and do not enter holdout estimates.

Five A/A pairs: fresh-process runtime variability

Recorded min 0.962508; max 1.046541; range 0.084033.



One training point and seed; excluded from selection and holdout. Shading is observed range, not a CI.
Source: native noise_summary.json · run-20261012T1543Z · recorded values only

All five A/A process ratios: native median 0.995031, min 0.962508, max 1.046541, range 0.084033. Shading is observed range, not a CI. [N06]

Kernel / same Fold64-DP6 policy	Verified cases	Mean process s	Mean online ms
barrett-L64-DP6-E0	9/9	2.923128	2683.602506
reference-L64-DP6-E0	9/9	2.903055	2674.156736

All nine arithmetic pairs recover the same answers and agree on exported outcomes, walk counters, policy and charges. This is recorded-semantic agreement, not a complete per-step trace. Exact Barrett mean process cost is 2.923128 s versus reference 2.903055 s, a neutral/slower observation within the measured A/A spread. The native rule retains reference arithmetic; no faster scalar-kernel claim is made.

Internal per-lane capped-walk restarts occur and remain in exported counters and charges. Their work is included in solve time. All completed-process statements refer to no process timeout or total step-cap failure; they do not assert zero internal restarts.

Sources: N01, N03, N06.

SECTION 05

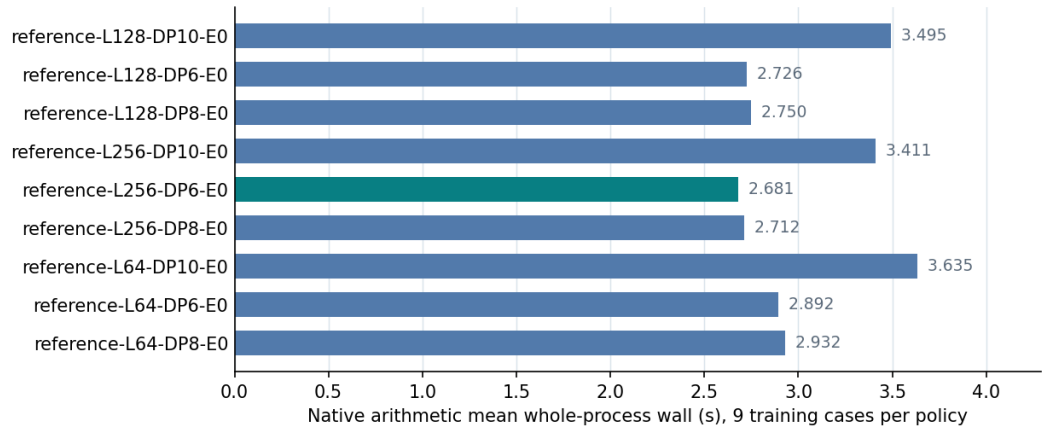
Every registered policy, including slower cases

The chosen scalar kernel receives all nine policies: lanes 64/128/256 × distinguished-point bits 6/8/10, with fruitless escape fixed off. Each policy sees the same three training points and three seeds per point. Lane count denotes lockstep work within one OS thread. A wider policy may change a walk trajectory and is judged by its complete process cost.

Policy	Verified / eligible	Mean process s	Mean online ms
reference-L128-DP10-E0	9/9; eligible	3.494705	3272.328770
reference-L128-DP6-E0	9/9; eligible	2.726115	2503.266741
reference-L128-DP8-E0	9/9; eligible	2.749576	2521.028409
reference-L256-DP10-E0	9/9; eligible	3.411012	3191.230668
reference-L256-DP6-E0	9/9; eligible	2.680834	2456.858188
reference-L256-DP8-E0	9/9; eligible	2.712132	2477.610694
reference-L64-DP10-E0	9/9; eligible	3.634782	3413.376393
reference-L64-DP6-E0	9/9; eligible	2.891766	2665.835796
reference-L64-DP8-E0	9/9; eligible	2.931672	2706.288331

Expanded lane / distinguished-point grid: every registered score

Lane counts 64 / 128 / 256 × DP6 / DP8 / DP10; selected scalar kernel; escape fixed off.



Teal marks the frozen native selection. Walk policies may change trajectories; total process cost is scored.

Source: native policy_selection.json · run-20261012T1543Z · recorded values only

Native registered training scores; teal marks the immutable native selection. All nine policies remain visible. This grid is not a global optimum. [N04]

Frozen native selection: reference-L256-DP6-E0, mean training process 2.680834 s; 64/DP6 scores 2.891766 s and 128/DP6 2.726115 s. All nine policies are eligible and all 81 processes verify within time and total-step limits. Selection precedes holdout. The default 64/DP6 is preferred only within 2% of the minimum.

Sources: N01, N04, N09.

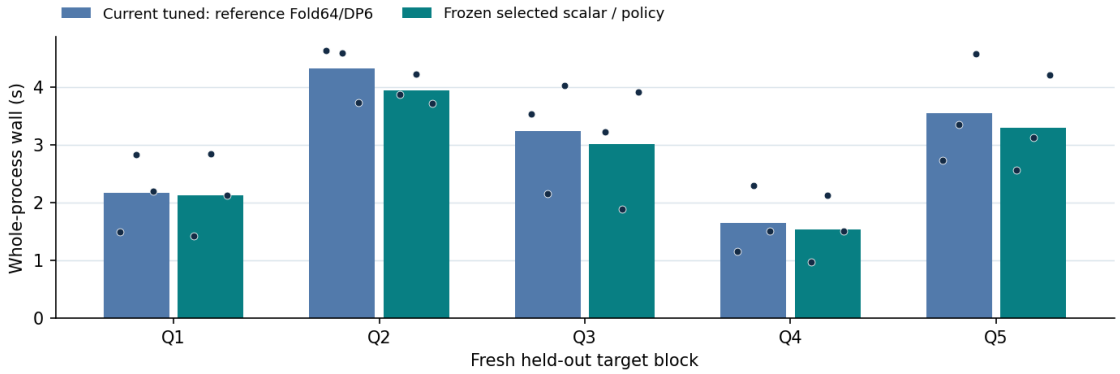
SECTION 06

Fresh held-out targets and native uncertainty

The primary estimate first takes each target's three-seed arithmetic process mean for each arm, then computes the geometric mean of the five current/selected ratios. Native SplitMix64 target-block bootstrap uses 20,000 paired resamples, seed 202610126299, with percentile 95% intervals. Three seeds nested in five targets do not become fifteen independent target samples.

Fresh holdout: all process times and native three-seed means

Five disjoint target blocks; dots retain all 30 individual solves. Bars are recorded native target means.



Both arms share one reference-scalar monomorphization and executable; only lane count differs.

Source: native statistics.json · run-20261012T1543Z · recorded values only

All 30 fresh held-out process costs and the native per-target three-seed means. Identical point/seed pairs alternate execution order. No fastest seed or target is removed. [N05]

Target	Current seed costs (s)	Selected seed costs (s)	Native means (s) / ratio
Q1	1.493073 2.833963 2.194318	1.417215 2.850072 2.118903	2.173785 / 2.128730 ratio 1.021165
Q2	4.638999 4.591936 3.734425	3.874468 4.221322 3.716998	4.321786 / 3.937596 ratio 1.097570
Q3	3.528888 2.154345 4.033394	3.225827 1.888357 3.908533	3.238876 / 3.007573 ratio 1.076907
Q4	1.152991 2.290008 1.501968	0.963513 2.126395 1.499933	1.648322 / 1.529947 ratio 1.077373
Q5	2.729404 3.356201 4.571203	2.563237 3.119802 4.213605	3.552269 / 3.298881 ratio 1.076810

All 30 cases independently verify, with no process timeouts or wrong answers. Seed triples are 202610126241–243, 244–246, 247–249, 250–252 and 253–255 for Q1–Q5. All individual process times remain above; native means use three paired seeds per target. Full rows retain online, CPU, RSS and counters.

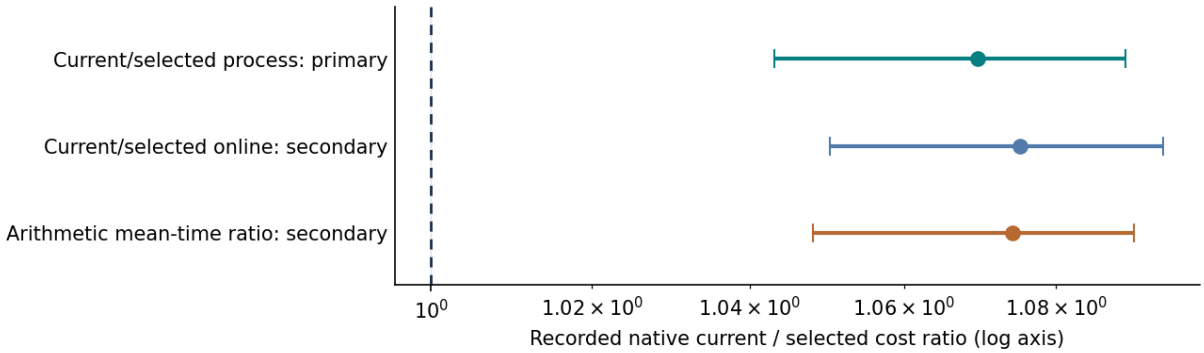
Sources: N05, N07, N09, N11.

SECTION 07

Timing boundary, host and supported interpretation

Native held-out ratios and 95% target-block intervals

Five paired target blocks; 20,000 native resamples preserve each target's three-seed means.



Values above 1 favor selected. Geometric and arithmetic estimands retain separate definitions.

Source: native statistics.json · run-20261012T1543Z · recorded values only

Native current/selected ratio direction. Values above 1 favor selected. Primary geometric and secondary arithmetic ratios preserve distinct estimands. The renderer does not estimate intervals. [N05]

Whole-process wall includes launch and output, point parsing, normal-basis conversion and target-independent setup, target-dependent walk, collision recovery and producer scalar replay. Online timing is secondary and excludes the recorded setup boundary. Independent C++ answer replay is an external correctness gate; its own recorded cost is reported separately.

Envelope / admission field	Actual value
Host and resources	AMD EPYC 9V74 model; KVM guest with 5 logical CPUs, cgroup quota 4 CPUs and memory 16 GiB. Rust 1.99.0; Cargo default release.
Pin / thread / address-space / timeout	CPU3; one OS thread; 12 GiB RLIMIT_AS; 60 s per rho child.
Monitor and reservation	Native 5 ms nominal wait4 poll plus scheduling overhead; all four campaign reservations and successful external replay uncontended.
Recorded campaign costs	423.474951 s actual child wall; 422.987101 s child CPU across all 139 runs. Excludes builds, replay and parent orchestration.
Independent replay cost	0.886574336 s wall; 0.833018 s user + 0.051324 s system; peak RSS 109,188 KiB. Outside solver intervals.
Unmet scientific gates	No new calibrated primitive-cost S; no L2 admission; no independent-host replay or observable physical-host exclusivity.

The complete uncontended panel supports a conditional host runtime improvement for the selected policy: 1.069651 [1.043178, 1.089238]. The lower interval edge overlaps the descriptive A/A range, and only five targets are sampled under exclusions. Secondary mean online costs are current 2755.666121 ms versus selected 2551.759882 ms. No generic boundary, exponent or historical IC/rho ratio changes.

Sources: N01, N05, N06, N11, N12.

SECTION 08

Validation, requirement status and reproduction

Requested obligation	Status and evidence / remaining gap
Faster optimized-control comparison	Verified on the declared host/panel: 139 runs retained; matched current tuned reference, fresh five-target holdout; conditional 1.069651 process ratio.
Exact scalar arithmetic	Implementation and finite verification complete; four Lean mathematical declarations checked. Rust machine-word bridges remain partial. Candidate is not selected.
Fresh targets and independent replay	Verified: 8 fresh points and exclusions, all 139 answers, 9 arithmetic semantic pairs; 0 invalid or globally capped solver processes.
Full required implementation checks	Full release library: 3,992 pass, 0 fail, 112 ignored; new worker: 7 pass; native runner: 22 pass. Initial compile and replay-wrapper failures preserved.
Comprehensive PDF and visuals	New source-linked addendum, combined edition and SVG/150 dpi PNG records; document-layout and full visual QA recorded beside report.
Calibrated S / independent-host admission	Partial: missing accounting and host admission bridges remain explicit; no ledger promotion.

Sources and raw records are authoritative; benchmarks holds identical browsing copies. This addendum preserves the broad basis/invariant synthesis and the two prior matched degree-83 studies as a frozen 60-page appendix.

Source	Evidence pointer
N01	Frozen prospective protocol
N02	New exact-scalar worker
N03	Arithmetic selection and all pairs
N04	All policy scores and frozen selection
N05	Native held-out statistics
N06	Native A/A noise summary
N07	Independent all-answer replay
N08	Fixture proposal/exclusion receipt
N09	Native runner and statistics source
N10	Arithmetic audit and proof status
N11	Root measurement review
N12	Host and frozen provenance

Pre-launch freeze commit adddc3555d3355f5d5d83fbf45cc6793cb6ada8f. The native file audit passes: 139 execution/producer/observation/environment records, 2,439 recorded hash checks, 455 distinct bound files and 1,036 snapshots. Frozen provenance and manifest bind the sources and receipts; no historical denominator is reused.

Conductor was attempted before edits and is unavailable. No chat transcript or credential content is included in the evidence.